

## **Lab Assignment: Case Study on Use Case Diagram**

Useful References -

1. What is a use case diagram? - [Use case diagram](#)
2. [Use case diagram explanation](#)
3. [Examples](#)

Tools - [StarUML](#) , [draw.io](#), [SmartDraw](#)

### **Case Study 1: Library Management System**

- a. Prepare a detailed use case diagram for a Library Management System, including all actors, use cases, and their interactions (associations, include and extend relationships) as described by the case study given below.
- b. Identify the two major use cases and write the use case specifications for the same

#### **Case Study Description:**

Design a Library Management System to simplify and automate how members, librarians, and administrators interact with library resources. The primary goal of the system is to ensure efficient management of books, users, and transactions such as issuing and returning books, while maintaining accurate records and minimizing manual effort.

The system should support different user types: Members, who primarily search, borrow, and return books; Librarians, who manage books, monitor transactions, and calculate fines and Administrators, who oversee user management and overall system operations. Each user interacts with the system through authentication mechanisms such as registration and login, and their access to functionalities depends on their assigned role.

Members can search for books using various criteria such as title, author, or category and view detailed information including availability status. Before a book is issued, the system must always verify the availability of the selected book to ensure that only available books can be borrowed. Members can return borrowed books within the due date, and during every return operation, the availability status of the returned book in the library catalog must be updated.

Librarians are responsible for maintaining the library catalog. They can add new books, update book information, remove damaged or outdated books, and manage member records. In case of overdue returns, librarians calculate the applicable fine based on library policies, and members can subsequently view and pay the fine through the system.

Administrators have higher-level control over the system. They can manage librarian accounts, generate usage reports, monitor system performance, and ensure smooth operation of all library activities.

## Case Study 2: Online Hospital Appointment & Care Management

- a. Prepare a detailed use case diagram for Online Hospital Appointment & Care Management System, including all actors, use cases, and their interactions (associations, include and extend relationships) as described by the case study given below.
- b. Identify the two major use cases and write the use case specifications for the same

### Case Study Description:

Imagine a hospital where patients no longer need to stand in long queues just to book an appointment or collect reports. The **Online Hospital Appointment & Care Management System** provides a digital platform that connects patients, doctors, hospital staff, and administrators in one integrated system.

Patients can either log in using an existing account or register as new users. Once logged in, they can search for doctors based on specialization, availability, or department. Patients can book appointments at their preferred time slots, reschedule or cancel appointments if their plans change, and view their appointment history. After consultation, patients can view prescriptions, download medical reports, and raise queries or complaints if issues arise regarding services or billing.

Doctors use the system to manage their daily schedules, view upcoming appointments, access patient medical history, and update diagnoses and prescriptions after consultations. Lab technicians upload test results, which are securely linked to patient profiles.

Hospital administrators ensure smooth operation behind the scenes. They manage doctor onboarding, department creation, appointment slots, user accounts, and monitor complaints. The system also integrates with an external notification service to send appointment reminders, prescription alerts, and report availability updates to patients.

Overall, the system improves efficiency, reduces waiting time, and ensures seamless communication between patients and hospital staff.